Solar Power Bank — Product Research

HOW DOES IT WORK? (process, method, outcome)

A 20,000mAh lithium-polymer battery sits inside a rugged, rubber-bumpered shell. You charge it from the wall through USB-C, which is the fast way, and the solar panel on the back trickle-charges it whenever it sits in sunlight. To charge devices, you lay a phone on the Qi wireless pad or plug into the USB-A and USB-C ports, up to four devices at once. LEDs show how much charge is left, and a built-in LED flashlight doubles as a torch during a blackout or at camp. The outcome is a phone-sized backup battery that stays ready in the car, the camping bag or the kitchen drawer, so a dead phone never leaves you stranded or unreachable.

Honest numbers (use these, not competitor claims):

- 20,000mAh is about 74Wh, enough for roughly 4 full phone charges.
- A built-in panel this size adds only a small amount per sunny day. Owners report a full day of direct sun adds roughly 20–25% to the bank, about one phone top-up.
- Solar is a backup top-up, not the main way to charge it.

Sixth-grade version It's a big battery for your phone with a small solar panel on the back. Charge it at home before you go. Then you can charge your phone by plugging it in or just laying it on top. If it sits in the sun, it slowly fills back up. It has a torch for when the power goes out or you're camping.

PRODUCT FEATURES

- 20,000mAh lithium-polymer battery (about 74Wh)
- Built-in solar panel on the back (no one states the wattage, so confirm it with the supplier)
- Qi wireless charging pad, up to 10W (auto-steps 10W / 7.5W / 5W)
- Wired outputs: 2× USB-A and 1× USB-C. The USB-C is two-way, charging the bank and devices through the same port.
- Fast-charge claims vary: 18W PD + QC3.0 (NOVA) vs "up to 20W" (Forejourney). Confirm with the supplier.
- Charges up to 4 devices at once
- Built-in LED flashlight (NOVA says dual)
- LED battery-level indicator
- Rugged ABS+PC shell with rubberised bumpers (NOVA adds "flame-retardant")
- Water resistance: the supplier says "waterproof", NOVA says IPX5 and Forejourney says "splash-resistant". Treat it as splash-resistant until it's certified.
- Compass carabiner (in NOVA's version)

- In the box: USB cable and manual

Confirm with the supplier before writing copy: solar panel output (W/mA), tested real capacity, IP rating certificate, wired output wattage, weight and dimensions, and battery certifications (UN38.3, MSDS, CE/FCC, and anything needed for the AU market).

STEP 4 — ASSUMED BENEFITS (what each feature does)

- 20,000mAh battery → about 4 phone charges, so you get through a weekend away or a long blackout without hunting for a plug
- Solar panel → tops itself up in the sun, so the bank doesn't sit flat in a drawer or glovebox when you finally need it
- Wireless pad → lay the phone down with no cable; charges earbuds and other Qi devices too
- 4 outputs at once → the whole family or group charges off one unit
- Flashlight → power and light in one grab-and-go item when the lights go out
- Rugged bumpered shell → survives being thrown in a car, tent, toolbox or backpack
- Battery-level LEDs → you know at a glance whether it's ready
- Two-way USB-C → one cable charges both the bank and your phone

Review-surfaced benefits the brand didn't lead with:

- It "lives in the car" as a permanent backup in case the car or phone dies.
- Solar works best as a trickle charger that stops the bank from draining while stored, not as a main charger.
- Wireless is handy for small devices like earbud cases while the phone charges by cable.

STEP 5 — ASSUMED DESIRES (ask "why?" until you hit the real want)

Master desire: "I never want to be stuck with a dead phone when it actually matters, like when the power's out, when we're off-grid, or when my family needs to reach me."

Primary Big-3 category: Health (safety and security). Secondary: Relationships (staying reachable for family).

- Charges phone 4× → Why? So I'm not cut off when there's no power → "I want to know I can always call for help or check on my kids."
- Solar top-up → Why? So it's actually charged when an emergency hits → "I want to be the one who's prepared, not the one scrambling."
- Flashlight + power in one → Why? Blackouts are stressful in the dark → "When the lights go out, I want my family calm, not panicking."

- Charges 4 devices → Why? Everyone's phone dies at camp → "I want the trip to be about the trip, not about whose phone is on 3%."
- Rugged build → Why? Cheap banks break outdoors → "I want gear I can chuck in the ute and forget about until I need it."

Deeper desires to test (Evolve Desire Calendar lens):

1. **Storm and blackout season** (Nov–Apr in QLD and northern AU; Jun–Nov hurricane season in the US): "When the power goes out, I want us connected and in control." This is in season right now.
2. **Summer holiday camping and road trips** (Dec–Jan in AU): "I want to switch off from the world without being unreachable."
3. **Christmas gifting** (Nov–Dec): "I want a gift for Dad or my partner that he'll actually use, not one that goes in a drawer."
4. **Travel and festivals:** "I want to stay charged all weekend without hunting for a powerpoint."

Desires to avoid chasing (the product can't credibly live up to them, per Evolve 1.1):

- "Never plug it in again" or "unlimited free energy from the sun"
- Fast solar charging ("full in a few hours of sun")
- Powering laptops, fridges or appliances. It's a phone-class bank, not a power station.
- "Fully waterproof / drop it in a river" unless it's certified IP67

STEP 6 — MECHANISM AND USP

Stated mechanism (how competitors say it works): "Harnesses the sun's energy" or "power anywhere", plus a feature list (solar + USB input, wireless + USB output). Nobody explains how the solar actually helps, and Forejourney inflates the capacity to 42,800mAh in places.

Assumed new mechanisms / USP candidates

- **"Sun-topped backup"** (solar as maintenance, not main charge) → Reframes the category's #1 complaint into an honest, believable benefit. No competitor says it. → **Strength: High**
- **Honest numbers** (about 4 phone charges; a day of sun ≈ one top-up) → New information in a market full of fake mAh and "unlimited" claims → **Strength: High**
- **Grab-and-go blackout kit** (power + torch + wireless in one) → Category-standard parts, but nobody packages them as a blackout kit → **Strength: Medium-High**
- **4 devices at once** → Common across the category → **Strength: Low-Medium**

Recommended one-line USP (assumed): "The backup battery that doesn't go flat in your drawer. Charge it at the wall, and the sun keeps it topped up for when the power isn't there."

Hidden mechanism the brand under-uses: Every lithium power bank slowly loses charge while stored, which is why the "emergency" bank is usually flat in an emergency. The panel's real job is to offset that drain and top it up on a sunny windowsill. Don't suggest leaving it on a hot car dashboard, because heat is bad for lithium batteries.

STEP 7 — HIDDEN MECHANISM ANALYSIS

1. What does the product do that it doesn't advertise?

- It works as a blackout torch and power source in one.
- It charges earbuds or a watch wirelessly while the phone charges by cable.
- It stays topped up in storage, so it's ready when you need it.
- One USB-C cable charges both ways.

2. What components does it include that competitors don't?

- Nothing physical. It's the same white-label unit. Win through the kit (carabiner, short cable set, pouch), a published tested-capacity spec sheet, certifications, warranty and local shipping.

3. If it worked too well, what would customers complain about?

- "Everyone at camp keeps borrowing it." "I had to buy a second one for the other car." "The kids took it."

4. If a customer explained it to a friend, what would they say? "It's a big power bank with a solar panel on it. Charge it at home first, because the sun part is slow, but it's a great backup to keep in the car or for when the power goes out. It's got a torch, and you can just sit your phone on it."

What that explanation reveals:

- **Buying trigger:** a trip or an outage.
- **Objection:** slow solar.
- **Outcome:** peace of mind as a backup that's always ready.

MARKET AWARENESS, SOPHISTICATION AND BELIEF

Awareness (assumed, split by sub-avatar)

- **Storm-season parent** → Problem aware → "Blackouts happen; I'll sort it when the next storm comes." → Enter on the problem: the last outage, phone on 8%, kids in the dark.
- **Camper / 4WD / road-tripper** → Solution aware → "Solar power banks are gimmicks." → Enter with new information: honest numbers and how to actually use it.

- **TikTok-exposed shopper** → Product aware → "It's $29.99 on TikTok." → Enter on the offer: bundle, warranty, local shipping, gift-ready.
- **Christmas gift buyer** → Aware of the need (a gift), not the product → "What do I get the guy who has everything?" → Enter on identity: "for the one who's always prepared."

Sophistication

Market stage: 4–5. Claims are already inflated (42,800mAh on a 20,000mAh unit, "unlimited energy"), and review sites openly call most solar power banks gimmicks. Bigger claims won't land.

- **New mechanism:** the "sun-topped backup" (solar keeps it ready, it doesn't replace the wall).
- **New information:** real numbers: about 4 phone charges, a day of sun ≈ one top-up, roughly 74Wh, which sits under the common 100Wh carry-on limit (carry-on only; check your airline).
- **New identity:** "the prepared one," the person in the family or group who always has power and a torch.

Belief barriers (Evolve module 9)

1. **Mechanism:** "Solar on a power bank doesn't actually work." → Agree, then reframe it as a top-up, and show a real sun test.
2. **Product:** "It's the same cheap Alibaba thing and the capacity is fake." → Tested capacity, certifications, warranty, real reviews.
3. **Self:** "I'm not a camper or a prepper; I don't need this." → Everyday use cases: glovebox, blackout drawer, school pick-ups, sport weekends.

VOICE OF CUSTOMER AND OBJECTIONS

Star breakdown

Not gathered yet. The only aggregates so far:

- RETMSR: 4.5★ from about 1.0K reviews
- Forejourney: claims 4.78★ (144), but its header says 15 reviews and no individual reviews show
- NOVA: no reviews

Verbatim language snippets

Sources: Amazon reviews of similar solar banks, via reseller listings and review-analysis sites.

- "Solar charging works, is excruciatingly slow (days/weeks)"

- "It seems to take FOREVER to charge with sunlight"
- "too small to do meaningful charging on an empty bank"
- "It lives in the car with me"
- "Solar panel actually charges at 300-400mah on a sunny day"
- "the wireless charging is very very slow"

Buzzwords to track: power outage, blackout, backup, emergency, peace of mind, never stranded, lives in my car / glovebox, off-grid, camping, rugged, trickle charge, real capacity, heavy / bulky, torch / flashlight.

Objection catalogue

- **"Solar is too slow"** → The most common complaint across reviews → Own it: "charge at the wall, the sun keeps it topped up," and show real numbers.
- **"The capacity is fake"** → Competitors claim 42,800mAh on a 20,000mAh unit → Publish the tested capacity and the number of phone charges.
- **"Too heavy / bulky"** → "Heavier than I expected" shows up in reviews → State the weight, and frame it as a glovebox or pack item, not a pocket item.
- **"Wireless is slow"** → Reviews → Set expectations: wireless for convenience and earbuds, USB-C for speed.
- **"Why pay more than $29.99 on TikTok?"** → RETMSR's price → Bundle (2-pack: car + home), warranty, local shipping, gift-ready packaging.
- **"Is it waterproof?"** → Specs conflict across sellers → Only claim the tested rating.
- **"Can I fly with it?"** → About 74Wh → Carry-on only; check your airline's rules.
- **"Can I leave it in the car?"** → Lithium and heat → Glovebox or shade, never a hot dashboard.

Gap

- No star breakdown or deep review set yet. Export reviews for a matching listing (RETMSR, BLAVOR or Riapow), the same process we used for FetchCam.
- Reddit research: r/camping, r/preppers, r/4x4australia, r/CampingandHiking, r/BuyItForLife, r/onebag.
- Order a sample and test it: real capacity, solar mA in Brisbane sun, and how much a full sunny day adds.